

B.Tech CSE Placement Roadmap

Personalized skill, certification & DSA roadmap | Dayananda Sagar University | Batch 2025-2028 | Target: Product company placements (6th semester onwards)

CURRENT SKILL SNAPSHOT (as of 2nd year)

Area	Current Status
Java	Learning basics - currently at loop-based pattern printing
DSA	Theory known (LinkedList, Stack, Queue, Graphs) but cannot yet solve LeetCode problems
MERN / Full Stack	Basic, explainable knowledge (HTML, CSS, JS, MongoDB) from coursework
Projects	Multiple vibe-coded projects with good ideas, but tech stack/architecture not understood
Tools used	Antigravity, Cursor, Google AI Studio, Firebase, Vercel
Git/GitHub	Knows add, commit, push only - no branching/merge/PR experience
Interest areas	Cybersecurity, Cloud Computing, Networking

GOLDEN RULE FOR THIS ROADMAP: Do not build the resume yet. Build real, explainable capability first (Java -> DSA -> understand your own projects -> Git -> Cloud/Security basics). The resume gets written AFTER substance exists, not before.

OVERALL TIMELINE OVERVIEW

Phase	Timeframe	Focus
Phase 1: Foundation	Now - Month 3	Java fundamentals, Git basics, project deep-dive, start DSA
Phase 2: Build Momentum	Month 3 - Month 7	Daily DSA practice, AWS cert, TryHackMe path, deepen MERN
Phase 3: Interview Mode	Month 7 - Month 10 (early 3rd year)	Mock interviews, system design basics, polish 2-3 projects, first resume draft
Phase 4: Placement Season	6th Semester onwards (~Month 11+)	Apply, interview, tailor resumes per role, keep DSA sharp

PHASE 1 - FOUNDATION (Now -> Month 3)

STEP 1 | Finish Java Fundamentals

- Loops, arrays, strings, functions -> OOP (classes, objects, inheritance, polymorphism)
- Goal: stop at 'pattern printing' and reach comfortable, bug-free coding
- Timeframe: 3-4 focused weeks

STEP 2 | Start Structured DSA Practice

- Order: Arrays -> Strings -> Recursion/Math -> Linked List -> Stack/Queue -> Trees -> Graphs -> Basic DP
- Solve 3-5 EASY problems per topic before attempting mediums
- Daily consistency matters more than long binge sessions

STEP 3 | Deep-Dive ONE Existing Project

- Pick PrepFlow AI or PassMate (pick the most complete one)
- Go file-by-file: routing, state management, API calls, DB schema, auth flow
- Goal: explain the entire data flow end-to-end without looking at code

STEP 4 | Learn Git Properly

- Branching, merging, resolving conflicts, .gitignore, meaningful commit messages, PR workflow
- One focused weekend (4-6 hours) is enough to close this gap

PHASE 2 - BUILD MOMENTUM (Month 3 -> Month 7)

STEP 5 | Daily DSA - No Pauses

- Move from Easy -> Medium problems steadily
- Target by end of 3rd year: 250-300 problems solved with genuine understanding
- Focus on speed + clarity, not just completion count

STEP 6 | Deepen MERN Understanding

- Go beyond basic usage: REST API design principles, MongoDB schema design, React state management
- Optional supporting certificate: freeCodeCamp (JavaScript Algorithms / Responsive Web Design) - free, credible

STEP 7 | Cybersecurity Foundation (Free, Hands-on)

- Complete TryHackMe 'Pre Security' path, then 'Cyber Security 101' path
- Free, verifiable profile link to add to resume later
- Builds on your existing interest + your phishing-detection research paper

STEP 8 | Cloud Certification - AWS Cloud Practitioner

- Use free AWS Skill Builder material
- 4-6 weeks part-time prep, pairs naturally with your Firebase/Cloud Run experience
- Pick ONE cloud ecosystem - do not chase AWS + GCP + Azure simultaneously

PHASE 3 - INTERVIEW MODE (Month 7 -> Month 10)

STEP 9 | System Design Basics

- Learn fundamentals: scalability, load balancing, caching, basic database design tradeoffs
- No need for deep distributed-systems theory yet - just enough for SDE-fresher level questions

STEP 10 | Polish 2-3 Projects Only

- Pick your strongest 2-3 (e.g. PrepFlow AI, SafeRoute AI, Kairos)
- Add live deployment links, clean READMEs, and be ready to explain every architectural decision
- Stop creating new vibe-coded projects - depth over quantity

STEP 11 | HackerRank Skill Certifications (Optional Polish)

- Java, Problem Solving, SQL tracks - quick, free, verifiable
- Only do these once skills are genuinely solid, not to fake competence

STEP 12 | First Resume Draft

- Now substance exists - organize it into role-specific resume versions
- Build 2-3 variants: SDE/Backend, Full-Stack, AI/ML-adjacent
- Same projects, different bullet-point framing depending on the role

PHASE 4 - PLACEMENT SEASON (6th Semester Onwards)

STEP 13 | Tailor & Apply

- Use the right resume variant per company/role type
- Keep refining bullet points based on actual interview feedback

STEP 14 | Sustain DSA + Mock Interviews

- Keep solving problems daily even during interview season - skills fade fast under stress otherwise
- Do mock interviews (peers, seniors, or platforms) to get comfortable under pressure

SKILLS & CERTIFICATIONS - QUICK REFERENCE

Tier	Skill / Area	Certificate (if any)
1 - Mandatory	Java + DSA (arrays to graphs/DP)	None - practice only (LeetCode/GFG)
1 - Mandatory	Git & GitHub (branching, PR workflow)	None needed
2 - Solidify	MERN depth (REST API design, schema design)	freeCodeCamp (optional, free)
3 - Differentiator	Cloud Computing (AWS/GCP basics)	AWS Certified Cloud Practitioner

3 - Differentiator	Cybersecurity fundamentals	TryHackMe Pre Security + Cyber Security 101
3 - Optional later	Networking (OSI, TCP/IP, DNS)	Cisco CCNA - only if going deep into networking
4 - Polish	SQL	HackerRank SQL certification
4 - Polish	Java / Problem Solving badges	HackerRank Skill Certifications

AVOID: Generic bootcamp 'Full Stack Certification' badges, collecting AWS+GCP+Azure at once, listing DBMS/OS/CN as 'skills' on resume (they are interview-tested fundamentals, not resume bullet points), and starting new vibe-coded projects before understanding existing ones.